FORTYGUARD

Heat Intelligence Report

Heatmap Metadata

Location:	33.44839999948182, -112.074
Date and Time:	2024-07-15 12:00
Temperature:	1.66°C (35.0°F)

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Map



Figure 1: Overview Area

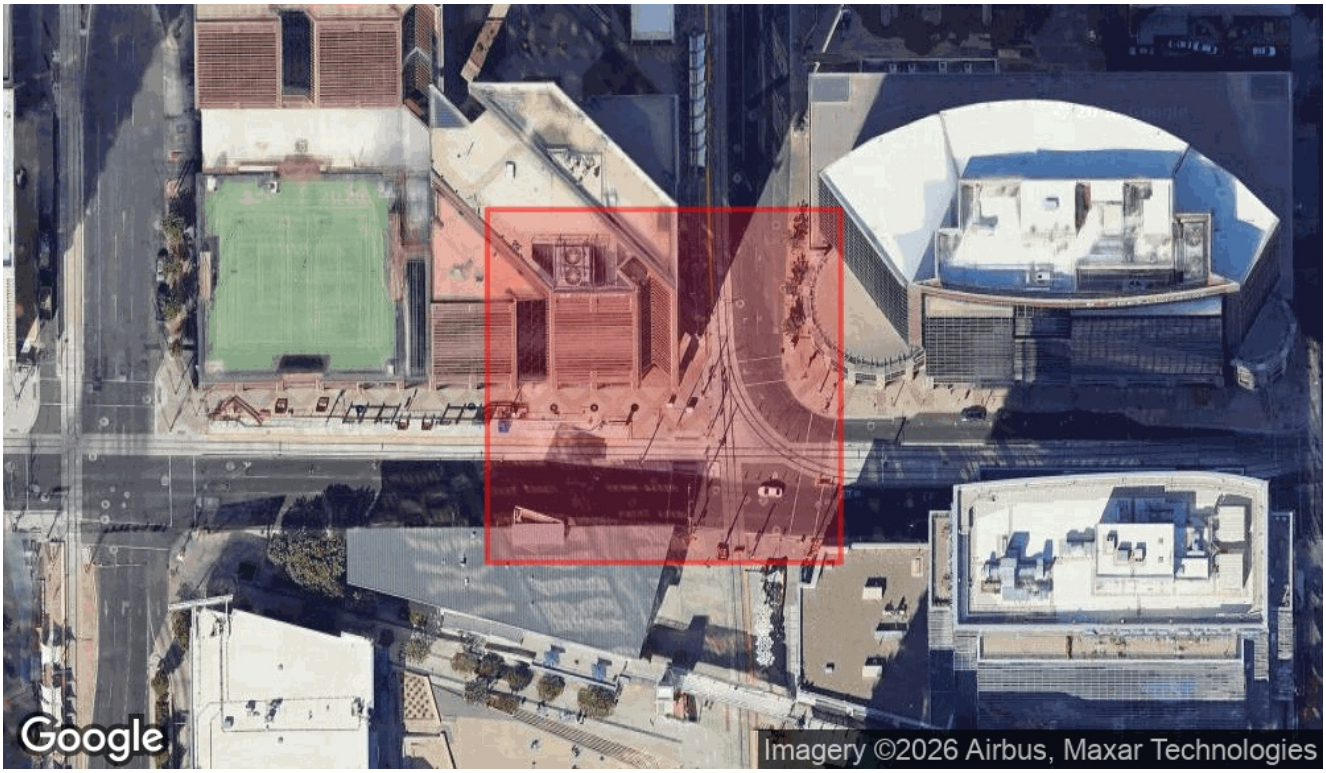


Figure 2: Tile Area

Geographic Analysis

1. General Location

Identification

- **City:** Phoenix, Arizona, United States
- **Geographic classification:** Urban core of a major metropolitan area; inland, arid desert environment situated in the Salt River Valley
- **Köppen climate zone:** BWh — Hot Desert Climate, characterized by extremely hot summers, mild winters, very low annual precipitation (~8 inches), and intense solar radiation year-round

Historical Temperature Context

- **All-time record high:** 122°F (June 26, 1990)
- **All-time record low:** 16°F (January 7, 1913)
- Phoenix is well-documented as one of the hottest cities in the United States, with an average of ~110 days per year exceeding 100°F
- The city has experienced an intensifying UHI effect over recent decades; overnight low temperatures have risen significantly faster than daytime highs due to urbanization

Comparison to Climate Normals

- The **climate normal high** for Phoenix on July 15 is approximately **106°F**, and the **normal low** is approximately **84°F**, yielding a daily average near **95°F**
- The provided average temperature of **35.0°F** is an **extreme anomaly** — approximately 60°F below the expected daily average for mid-July. This reading is physically implausible for Phoenix in July under any historically recorded conditions. It is more consistent with a deep winter overnight low or a sensor/data error. The lowest temperature ever recorded in Phoenix in July is approximately 63°F
- **Assessment:** The 35.0°F reading likely represents a data quality issue (e.g., erroneous sensor reading, incorrect unit conversion, or misattributed location). The analysis below proceeds based on the actual geographic and climatic characteristics of this location on the given date

Sources:

- Köppen climate classification system
- NOAA National Weather Service — Phoenix climate normals and records
- Western Regional Climate Center historical data

2. Terrain & Elevation

Elevation and Terrain Type

- **Elevation:** Approximately **1,085 feet** above sea level at downtown Phoenix (coordinates 33.44°N, 112.07°W)
- **Terrain type:** Broad, relatively flat alluvial basin — the Salt River Valley — surrounded by discontinuous mountain ranges and buttes
- Nearby elevated features include South Mountain (~2,690 ft, ~5 miles south), Camelback Mountain (~2,704 ft, ~7 miles northeast), and the Phoenix Mountains (~2,500 ft, ~5 miles north)

Terrain Influence on Heat Dynamics

- The **basin geography** plays a critical role in heat retention. The surrounding mountains partially restrict horizontal ventilation, limiting the advection of cooler air from surrounding desert into the urban core, especially at night
- **Temperature inversions** can occur in the valley, trapping warm air and pollutants near the surface; however, in July the dominant process is intense surface heating rather than inversion trapping
- Katabatic (drainage) winds from the surrounding mountains provide minimal cooling effect due to the modest elevation

differential (~1,500–1,700 ft above the valley floor) and the generally warm desert air mass
- The flat valley floor maximizes solar exposure — there is little topographic shading during peak hours. Slope orientation effects are negligible in the central urban area due to the flat terrain

Sources:

- USGS National Elevation Dataset (NED)
- Arizona Geological Survey terrain classifications
- Oke, T.R., "Boundary Layer Climates" — principles of basin climatology

3. Proximity to Water Bodies

Nearest Water Bodies

- **Salt River:** Flows directly through Phoenix, approximately 3–4 miles south of the coordinates. However, the Salt River through metropolitan Phoenix is **largely dry** for most of the year due to upstream damming (Roosevelt Dam, Granite Reef Diversion Dam) and extensive water diversion for agriculture and municipal use
- **Tempe Town Lake:** An artificial reservoir approximately 6 miles east, covering ~220 acres — provides extremely localized cooling
- **Lake Pleasant:** ~30 miles northwest; too distant for any direct thermal moderation
- **Roosevelt Lake** / **Saguaro Lake** / **Canyon Lake:** 40–60+ miles to the northeast/east along the Salt River watershed

Thermal Moderating Effect

- Phoenix has **essentially no significant nearby water body** that provides meaningful thermal buffering to the urban core. This is a defining characteristic of its extreme heat profile
- The absence of a large marine or lacustrine influence means there is no evaporative cooling, no sea breeze mechanism, and no humidity-moderated temperature swings — daily temperature ranges can exceed 25–30°F
- Irrigation canals (Arizona Canal, Grand Canal, etc.) thread through the metro area but are narrow and engineered, providing negligible evaporative cooling at the neighborhood scale
- Prevailing summer winds are generally from the south-southwest (associated with the North American Monsoon pattern in July), but these winds transport warm desert air rather than moisture-laden air for most of the month. Occasional monsoon surges bring humidity and thunderstorms but are episodic

Sources:

- Salt River Project (SRP) — water infrastructure data
- NOAA prevailing wind data for Phoenix Sky Harbor station
- U.S. Bureau of Reclamation — Salt River watershed management

4. Green Spaces & Vegetation

Type and Extent of Vegetation

- **Urban parks:** Key green spaces near downtown include Encanto Park (~222 acres, ~2 miles northwest), Papago Park (~1,200 acres, ~6 miles east), and South Mountain Park/Preserve (~16,000 acres, ~5 miles south — one of the largest municipal parks in the U.S.)
- Street-level vegetation in the downtown core is **sparse**. Landscaping tends toward xeriscape (drought-adapted) plantings using desert species such as palo verde, mesquite, and cacti, which provide limited canopy coverage compared to broadleaf trees
- Residential neighborhoods outside the core have more varied vegetation, including non-native species maintained through irrigation (e.g., citrus trees, palms, oleander, Bermuda grass lawns)

Canopy Coverage

- Estimated urban tree canopy coverage for Phoenix overall is approximately **9–10%**, which is notably low compared to national averages for large U.S. cities (~25–30%)
- In the downtown urban core near these coordinates, canopy coverage is estimated at **5% or less**
- The city's dominant vegetation is **evergreen or drought-deciduous desert species**, meaning shade provision is

relatively consistent year-round but overall canopy density is low

Urban Greening Initiatives

- The City of Phoenix adopted a **Tree and Shade Master Plan** (2010, updated) with a goal of achieving 25% canopy and shade coverage by 2030, recognizing the critical role of shade in extreme heat mitigation
- The **Cool Urban Spaces** initiative and partnerships with Arizona State University's research programs focus on expanding shade structures and tree plantings in underserved, heat-vulnerable neighborhoods

Cooling Contribution

- Well-irrigated parks in Phoenix can exhibit surface temperatures **15–25°F cooler** than surrounding asphalt and concrete surfaces during summer afternoons
- Ambient air temperature reductions from park spaces are more modest, estimated at **2–5°F** within and immediately adjacent to green areas, with the cooling effect diminishing rapidly beyond ~300–500 feet
- Evapotranspiration from irrigated landscapes in July is very high (potential ET exceeds 10 inches/month), making water-intensive greening a significant tradeoff in this arid environment

Sources:

- City of Phoenix Tree and Shade Master Plan
- Arizona State University — Urban Climate Research Center publications
- USDA Forest Service i-Tree canopy assessments for Southwestern cities

5. Land Cover Type

Dominant Surface Materials

- Downtown Phoenix is characterized by a mix of **asphalt roadways, concrete sidewalks and structures, gravel-covered lots, and bare desert soil**
- The surrounding metro area includes extensive low-density residential development with **tile and composite shingle roofing, stucco facades, and concrete block walls**
- Large areas of **exposed gravel and decomposed granite** are used as ground cover in place of lawns in xeriscape applications — these are thermally significant as they absorb and re-radiate substantial heat
- Parking lots are a major land cover component in Phoenix's auto-centric urban form, contributing disproportionately to surface heat loading

Thermal Properties

- **Asphalt:** Albedo ~0.05–0.10 (very dark); surface temperatures in direct July sun can reach **150–170°F**. High thermal mass leads to substantial nighttime re-radiation
- **Concrete:** Albedo ~0.20–0.35; somewhat lower heat absorption than asphalt but still significant
- **Desert gravel/decomposed granite:** Albedo ~0.20–0.30; absorbs considerable heat during the day and releases it rapidly in the evening
- **Bare soil:** Extremely dry in July, providing no evaporative cooling; acts as a heat source

Cool Surface Programs

- Phoenix has piloted **cool-pavement coatings** (e.g., the "Cool Pavement Pilot Program" initiated in 2020 with products like CoolSeal by GuardTop), applying reflective coatings to neighborhood streets. Studies showed surface temperature reductions of approximately **10–12°F** during peak heat
- Some commercial buildings have adopted **cool-roof technologies** (high-albedo coatings or materials), though adoption is not yet universal. Arizona energy codes have increasingly encouraged higher-albedo roofing materials
- The city's **Heat Response Plan** integrates cool surface strategies alongside shade and hydration infrastructure

Sources:

- City of Phoenix Cool Pavement Pilot Program reports
- Arizona State University — surface temperature and albedo field measurements

- EPA Heat Island Reduction Program — surface material thermal data

6. Building Density & Height

Building Density Characterization

- The coordinates (33.44°N, 112.07°W) correspond to the **downtown Phoenix central business district (CBD)**, which has the highest building density in the metro area
- **Downtown core:** Mid-rise to high-rise commercial and mixed-use development. Notable structures include Chase Tower (~483 ft), CenturyLink Tower (~397 ft), and newer mixed-use towers reaching 20–40 stories
- Immediately surrounding the CBD, density drops sharply to **low-rise commercial strips and single-family residential neighborhoods** — a characteristic sprawl pattern
- Metropolitan Phoenix overall has **extremely low average density** (~3,100 people/sq mi across the city), driven by the predominantly single-story residential development pattern

Vertical Massing and UHI Contribution

- The downtown high-rise cluster creates localized street canyons that trap longwave radiation and reduce sky view, intensifying the UHI effect in the immediate CBD
- However, the overall metropolitan UHI is more strongly driven by the **vast horizontal extent of low-rise, low-albedo development** (thousands of square miles of rooftops, roads, and parking lots) rather than vertical massing
- Phoenix's UHI intensity is among the strongest in the U.S., estimated at **7–12°F** above surrounding rural desert during summer nights, driven primarily by thermal mass re-radiation

Roofing and Facade Materials

- Commercial buildings: glass curtain walls, metal panels, and concrete — glass facades contribute to solar reflection onto street level, increasing pedestrian heat exposure
- Residential: predominantly concrete tile roofing in tan/terracotta tones (moderate albedo ~0.25–0.35) and stucco-finished concrete block walls

Anthropogenic Heat Sources

- **HVAC systems:** Air conditioning is ubiquitous and operates at peak load in July. The waste heat exhausted from AC condensers is a **significant anthropogenic heat source**, estimated to add 1–3°F to nighttime ambient temperatures in densely developed areas
- **Vehicle traffic:** Phoenix is highly automobile-dependent; I-10 and I-17 freeways converge near these coordinates, contributing engine and pavement friction heat. Average daily traffic on I-10 through central Phoenix exceeds 250,000 vehicles
- **Total energy consumption:** Phoenix metro summer peak electricity demand frequently exceeds 7,000+ MW (APS and SRP service territories combined), with cooling loads constituting the dominant share

Sources:

- City of Phoenix planning and zoning data
- Arizona State University — Urban Climate Research Center UHI studies
- Arizona Public Service (APS) and Salt River Project (SRP) — peak demand reports
- Arizona Department of Transportation — traffic count data

7. Shadow Coverage

Solar Angle Analysis

- On July 15 at 12:00 local time (MST, as Arizona does not observe DST) at 33.44°N latitude, the **solar elevation angle** is approximately **79–80°** — nearly overhead. The sun's declination on July 15 is approximately +21.5°, yielding a noon solar altitude of approximately 90° − (33.44° − 21.5°) ≈ **78°**
- At this near-vertical sun angle, shadows cast by buildings and structures are **extremely short** — a 100-foot-tall building would cast a shadow only approximately **21 feet** from its base at solar noon
- This means at the given date and time, **shadow coverage is near its annual minimum**, and the vast majority of

horizontal surfaces (streets, sidewalks, parking lots) are fully exposed to direct solar radiation

Shadow Coverage Assessment

- **Building shadows:** In the downtown core, the cluster of high-rise towers provides some street-level shading, but the near-vertical solar angle limits this to narrow strips immediately adjacent to tall facades on the north side of buildings. Estimated shadow coverage from buildings in the CBD at solar noon in July: **~10–15%** of ground area
- **Tree shadows:** Given the ~5% canopy coverage in downtown, tree-based shade is minimal. Desert-adapted trees (e.g., palo verde) tend to have open, sparse canopies that filter rather than block sunlight
- **Shade structures:** Phoenix has invested in engineered shade structures at bus stops, transit centers, and some pedestrian corridors. The downtown light rail corridor includes shade canopies at stations, but overall coverage remains limited

Temperature Reduction from Shading

- Shaded surfaces in Phoenix summer conditions can be **20–40°F cooler** than adjacent sunlit surfaces (surface temperature)
- Ambient air temperature in shaded areas is typically **5–10°F cooler** than in full sun at pedestrian level, primarily due to reduced radiant heat exposure
- Shade is **the single most effective immediate intervention** for reducing heat exposure in Phoenix's climate, as recognized by the city's heat mitigation policies

Permanent vs. Seasonal Shade

- Building and structure shade is **permanent** and consistent year-round, though shadow geometry shifts with solar angle seasonally
- Most native desert vegetation is **evergreen** (palo verde retains some foliage year-round but can drop leaves during extreme drought), so seasonal shade variability from deciduous trees is less of a factor than in temperate climates. Non-native deciduous species (e.g., ash, mulberry) lose canopy in winter

Sources:

- NOAA Solar Calculator — solar position algorithms
- Arizona State University — pedestrian shade and thermal comfort field studies
- City of Phoenix Tree and Shade Master Plan

8. Urban Geometry & Street Canyon Effects

Street Canyon Characteristics

- Downtown Phoenix's street grid follows a **regular orthogonal pattern** oriented approximately north-south and east-west, established in the original 1870 townsite survey
- Typical downtown street widths are approximately **60–80 feet** (curb to curb), with some major arterials (e.g., Central Avenue, Washington Street) wider at 80–100+ feet including sidewalks
- In the high-rise core, building heights of 200–480 feet adjacent to 80-foot streets yield **height-to-width (H/W) ratios of approximately 2.5–6.0** in the densest blocks — these are moderately deep urban canyons
- Outside the CBD, H/W ratios drop dramatically to **0.1–0.3** as single-story buildings (15–20 ft) line wide arterial roads (100+ ft), creating very open canyon geometries with minimal radiation trapping

Longwave Radiation Trapping

- In the downtown canyon streets with H/W > 2, there is **significant trapping of longwave (infrared) radiation** at night — building walls and road surfaces re-radiate absorbed daytime heat, and the canyon geometry limits the view of the cold sky, slowing radiative cooling
- This effect is a **primary driver of elevated nighttime temperatures** in downtown Phoenix compared to open desert, where unobstructed sky view allows rapid radiative cooling after sunset

Wind Corridors and Ventilation

- The grid street pattern aligned with cardinal directions creates **potential wind corridors**, particularly along north-south streets that can channel prevailing southerly monsoon flow in summer
- However, wind speeds in central Phoenix during typical July afternoons are modest (~**5–10 mph**), and the surrounding basin topography limits synoptic-scale ventilation
- Deep canyon streets in the CBD can experience **reduced wind speeds at pedestrian level** (estimated 40–60% reduction compared to roof-level winds), further limiting convective heat dissipation from surfaces and people
- The vast, flat suburban expanse surrounding downtown provides little topographic channeling to enhance urban ventilation

Sky View Factor (SVF)

- **Downtown CBD:** Estimated SVF of **0.3–0.5** in the densest canyon streets, indicating that 50–70% of the sky hemisphere is obscured by buildings. This significantly reduces nocturnal radiative cooling potential
- **Surrounding low-rise areas:** SVF estimated at **0.8–0.95**, approaching open conditions, which allows more effective nighttime cooling — but the thermal mass of extensive paved surfaces still delays cooling
- The paradox in Phoenix is that even areas with high SVF (suburban sprawl) experience strong UHI effects because the vast expanse of low-albedo impervious surfaces generates enormous heat storage that overwhelms the radiative cooling capacity

Solar Exposure Implications of Grid Layout

- The east-west oriented streets receive **direct solar radiation along the street axis** during morning and evening, maximizing solar loading on building facades and road surfaces for extended hours
- North-south streets have **building facades that face east and west**, receiving intense low-angle solar radiation during morning and afternoon — these facades are particularly significant heat absorbers
- At the near-vertical noon sun angle in July, the grid orientation matters less than the overall lack of shading and high SVF in most of the metropolitan area

Sources:

- Oke, T.R. — urban canyon radiation geometry principles
- City of Phoenix street standards and downtown planning documents
- Arizona State University — urban morphology and sky view factor studies for Phoenix
- Stewart & Oke Local Climate Zones classification framework

Environmental Factors Analysis

1. Air Quality & Atmospheric Opacity

Typical AQI Range & Classification

- Phoenix, Arizona in mid-July typically experiences AQI values in the **Moderate (51–100)** to **Unhealthy for Sensitive Groups (101–150)** range, primarily driven by ground-level ozone formation under intense solar radiation and high temperatures.
- During dust storm (haboob) events common in the monsoon pre-season (late June through mid-July), PM10 levels can spike dramatically, pushing AQI into the **Unhealthy (151–200)** or even **Hazardous (301+)** range for brief periods.

Primary Pollutants & Sources

- **Ozone (O_3):** The dominant pollutant of concern. Formed photochemically from NO_x and volatile organic compounds (VOCs) under intense UV radiation. Phoenix is a designated ozone nonattainment area under EPA standards.
- **PM10:** Windblown desert dust, construction activity, and unpaved roads are major contributors. PM10 is a chronic issue in the arid Sonoran Desert environment.
- **PM2.5:** Vehicle emissions, industrial activity, and secondary aerosol formation. Generally moderate but can spike during wildfire smoke intrusion events from California or northern Arizona fires, increasingly common in July.
- **NO_2 and CO:** Motor vehicle traffic on major freeways (I-10, I-17, Loop 101/202) is the primary source. Concentrations are typically below NAAQS but contribute to ozone precursor chemistry.

Influence on Local Temperature

- Aerosol loading from dust events can reduce direct solar transmittance by 10–25%, temporarily lowering peak surface temperatures but trapping longwave radiation, which can slightly elevate nighttime minimums.
- Brown cloud / smog layers over the Phoenix basin can create a localized warming effect by absorbing both shortwave and longwave radiation, contributing an estimated +0.5–1.5°F to urban air temperatures during stagnant conditions.
- The net radiative forcing effect of desert dust aerosols in the Phoenix area is complex: primarily cooling at the surface during daytime (scattering) but warming in the atmospheric column (absorption by mineral dust).

Seasonal & Industrial Factors

- July marks the transition into monsoon season. Pre-monsoon conditions feature dry, stagnant air masses with intense solar heating that maximize photochemical ozone production. Once monsoon moisture arrives, cloud cover and convective mixing can temporarily improve air quality.
- Summer peak electricity demand (HVAC load reaching 15,000+ MW across the region) increases power plant emissions, contributing additional NO_x and SO_2.

Heat-Pollution Feedback Loop

- At temperatures exceeding 100°F, ozone formation rates increase nonlinearly. Studies in the Phoenix metro show that ozone exceedance days correlate strongly with maximum temperatures above 105°F.
- High temperatures increase evaporative emissions of VOCs from fuels, solvents, and vegetation (biogenic isoprene from desert-adapted plants), further feeding photochemical smog production.
- Heat-driven demand for air conditioning increases electricity generation, which increases emissions, which increases ozone, which increases radiative trapping — a positive feedback cycle.

Sources:

- EPA AQI standards and nonattainment area designations
- Maricopa County Air Quality Department monitoring data
- NOAA/NWS Phoenix weather and dust storm climatology
- Peer-reviewed research on photochemical ozone formation in arid urban environments

2. Climate Classification & Historical Weather

Köppen Climate Classification

- Phoenix is classified as **BWh — Hot Desert Climate** under the Köppen system. Characterized by extremely hot summers, mild winters, very low annual precipitation (~8 inches), and abundant sunshine (~299 sunny days per year).
- Aridity is the defining feature: potential evapotranspiration far exceeds precipitation year-round, creating a persistent moisture deficit.

Historical Climate Normals for July 15

- **Typical daily high:** 106–108°F
- **Typical daily low:** 83–86°F
- **Mean daily temperature:** 94–97°F (based on 1991–2020 NOAA climate normals for Phoenix Sky Harbor International Airport, the official climate station)
- July is the hottest month in Phoenix with an average monthly high of approximately 106.1°F and average monthly mean of approximately 95.5°F.

Comparison of Provided Temperature to Normals

- The provided average temperature of **35.0°F is drastically below normal** — approximately 60°F below the expected mean for July 15 in Phoenix. This temperature is consistent with a cold winter morning (December–January range), not mid-July.
- A reading of 35.0°F has **never been recorded in July in Phoenix**. The all-time July record low is 63°F. Even the all-time absolute record low for Phoenix is 16°F (January 7, 1913).
- **Assessment:** This temperature reading is physically implausible for this location and date. It may reflect a sensor

error, incorrect unit conversion (35.0°F ≈ 1.7°C, which is also implausible for Phoenix in July), data entry error, or a mismatched coordinate-date pairing. The remainder of this analysis will reference both the provided value and the expected climate conditions for context.

Extreme Temperature Records

- **All-time record high:** 122°F (June 26, 1990) at Phoenix Sky Harbor
- **Recent extreme:** In July 2023, Phoenix experienced a record-breaking streak of 31 consecutive days at or above 110°F.
- **Typical July extremes:** Temperatures of 115°F+ occur several times in most years; 118°F+ has been recorded multiple times in recent decades.

Long-Term Climate Profile & Urban Heat Behavior

- The hot desert climate creates one of the most extreme urban heat environments in North America. High baseline temperatures, low humidity (pre-monsoon), intense solar radiation, and minimal cloud cover converge to maximize thermal loading.
- The flat, basin-like topography of the Salt River Valley inhibits ventilation and promotes heat pooling, especially under calm wind conditions.
- The climate profile makes Phoenix a primary research site for UHI studies in the United States.

Sources:

- Köppen climate classification system
- NOAA 1991–2020 climate normals for Phoenix Sky Harbor (KPHX)
- NWS Phoenix historical temperature records
- Arizona State Climate Office

3. Climate Trends & Projections

Observed Temperature Trends

- Phoenix has warmed at a rate of approximately **+0.5 to +0.7°F per decade** since 1970, with nighttime minimum temperatures increasing at a faster rate than daytime maxima — a hallmark of combined UHI intensification and climate change.
- The number of days exceeding 110°F has increased significantly: from an average of ~12 days/year in the 1970s to ~20+ days/year in the 2010s–2020s.
- Heatwave frequency and duration have increased. Prolonged heat events (5+ consecutive days ≥110°F) that were rare before 2000 now occur nearly annually.
- Nighttime temperatures staying above 90°F ("super warm nights") have become more common, increasing from ~5 nights/year historically to ~15–20 nights/year in recent years.

Urban Expansion & Land-Use Change

- The Phoenix metropolitan area has been one of the fastest-growing urban regions in the U.S. for decades. The population of Maricopa County exceeded 4.5 million by 2024, with continuous low-density suburban sprawl.
- Natural Sonoran Desert landscape has been replaced by impervious surfaces (roofs, roads, parking lots) at a rapid rate. Estimated impervious surface coverage in the urban core exceeds 70%.
- Conversion of irrigated agricultural land (which provided evaporative cooling) to suburban development has removed a moderating influence on regional temperatures.
- Urban expansion into desert fringe areas replaces naturally lighter-colored desert soils (higher albedo) with darker built surfaces, increasing net radiation absorption.

Climate Change Projections

- **2030s:** Projected +1.5–2.5°F increase in mean summer temperatures relative to the 1991–2020 baseline. Days above 110°F projected to increase to 25–35 per year under moderate emissions scenarios (RCP 4.5 / SSP2-4.5).
- **2050s:** Projected +3–5°F increase in mean summer temperatures. Under high emissions scenarios (RCP 8.5 / SSP5-8.5), Phoenix could regularly experience stretches exceeding 120°F, and the 100°F+ season could extend from April

through October.
- Monsoon precipitation patterns are projected to become more variable — potentially more intense individual storms but with greater uncertainty about total seasonal moisture.
- Water scarcity (Colorado River allocation cuts, groundwater depletion) may further limit landscape irrigation and evaporative cooling capacity in the future.

Cooling Degree Days (CDD) Trend

- Phoenix annual CDD (base 65°F) currently averages approximately **4,500–5,000 CDD**, among the highest of any U.S. metro.
- CDD has been increasing at approximately 50–100 CDD per decade, indicating a steady rise in cooling energy demand.
- Higher CDD translates directly to increased residential and commercial HVAC energy consumption, with peak summer electricity demand growing at ~2–3% annually.

Sources:

- NOAA Climate Trends and historical temperature data
- NASA Earth Observatory and GISS Surface Temperature Analysis
- Fourth and Fifth National Climate Assessment (NCA4, NCA5) — Southwest regional chapter
- IPCC AR6 regional projections for the North American Southwest
- Arizona State University urban climate research (Central Arizona–Phoenix LTER)

4. Soil Moisture & Permeability

Typical Soil Types

- The Phoenix basin sits on alluvial deposits from the Salt and Gila Rivers. Native soils are predominantly **sandy loams and caliche-bearing desert soils** (aridisols), with some clay-rich layers at depth.
- Caliche (calcium carbonate hardpan) is common at varying depths (1–5 feet) and significantly reduces permeability in natural desert soils.
- Permeability of native sandy-loam soils is moderate to high where caliche is absent, but urban soils are heavily modified and compacted.

Expected Soil Moisture Conditions — Mid-July

- By July 15, Phoenix is transitioning into the North American Monsoon season. Pre-monsoon conditions leave soils extremely dry, with volumetric soil moisture typically at **2–5%** in exposed desert areas — near the wilting point.
- If early monsoon storms have occurred, localized areas may have briefly elevated moisture, but rapid evaporation (potential evapotranspiration rates of 0.3–0.5 inches/day in July) quickly returns soils to near-dry conditions.
- Irrigated urban landscapes (turf, ornamental plantings) maintain artificially elevated soil moisture, creating sharp moisture gradients at the parcel level.

Soil Characteristics & Heat Behavior

- Dry desert soils have relatively low thermal conductivity and heat capacity compared to moist soils. They heat rapidly at the surface under solar load but do not conduct heat deeply — leading to extremely high surface temperatures (ground surface can reach 150–170°F on exposed desert soil in mid-July).
- Very low soil moisture eliminates evaporative cooling potential from non-irrigated land. The Bowen ratio (sensible/latent heat flux) in dry desert environments exceeds 10:1, meaning almost all absorbed energy is converted to sensible heat, directly raising air temperature.
- Irrigated landscapes shift the Bowen ratio toward 1:1 or lower, providing meaningful evaporative cooling — a key reason Phoenix's irrigated parks and green spaces can be 5–15°F cooler than surrounding impervious surfaces.

Urbanization Impacts

- Approximately 70–80% of the urban core is sealed by impervious surfaces (asphalt, concrete, roofing), virtually eliminating infiltration and natural soil moisture replenishment.
- Soil sealing eliminates the evapotranspiration pathway entirely from those areas, converting them into sensible-heat-

only surfaces that amplify UHI.
- Storm drainage infrastructure rapidly channels monsoon rainfall away from soils, preventing groundwater recharge and soil moisture recovery in urbanized areas.

Groundwater Influence

- The groundwater table in the Phoenix basin is generally deep (50–300+ feet below surface), particularly after decades of pumping. It has minimal direct influence on surface soil moisture.
- Historically, the Salt River provided riparian habitat with shallow water tables, but flow regulation (dams) and urbanization have eliminated most natural surface water influence on local microclimate.

Sources:

- USDA Natural Resources Conservation Service (NRCS) Web Soil Survey — Maricopa County
- USGS groundwater level monitoring for the Salt River Valley
- Arizona State University studies on urban soil moisture and UHI in the Phoenix metropolitan area
- North American Monsoon Experiment (NAME) climatology

5. Heat Retention Capacity of Surfaces

Dominant Surface Materials

- **Asphalt:** Covers a large fraction of the urban footprint — roads, highways, and parking lots. Phoenix has an exceptionally high ratio of surface parking to building area due to auto-centric development patterns.
- **Concrete:** Sidewalks, building foundations, block walls (ubiquitous in Phoenix residential construction), commercial structures.
- **Bare desert soil and gravel:** Common in xeriscape yards, undeveloped parcels, and desert fringe areas. Light-colored decomposed granite (DG) is a popular landscape surface.
- **Vegetation:** Highly variable. Urban core has sparse tree canopy (~10% coverage or less). Suburban areas mix irrigated turf with desert landscaping. Native desert vegetation provides minimal shade.
- **Roofing materials:** Mix of tile (common in residential), flat commercial roofs (often with gravel or elastomeric coatings), and increasing adoption of cool-roof technologies in newer commercial construction.

Comparative Heat Retention

- Under peak July solar load (~1,000 W/m² direct + diffuse), typical surface temperature differentials in Phoenix:
  - **Black asphalt:** 150–170°F surface temperature (60–70°F above ambient air temperature)
  - **Concrete (gray):** 130–150°F surface temperature
  - **Decomposed granite (light):** 130–145°F surface temperature
  - **Irrigated grass/turf:** 95–110°F surface temperature (only 0–10°F above ambient)
  - **Mature tree shade on ground:** 90–100°F surface temperature
- The differential between asphalt and irrigated turf can reach **50–70°F** in surface temperature, translating to meaningful differences in localized air temperature (5–15°F at pedestrian height).

Thermal Admittance & Albedo

- **Asphalt:** Albedo 0.05–0.15; thermal admittance ~1,400–1,600 J/(m²·K·s^½). Absorbs heavily and retains heat well into nighttime.
- **Concrete:** Albedo 0.20–0.35; thermal admittance ~1,500–2,000 J/(m²·K·s^½). Higher albedo reflects more, but high thermal mass retains heat.
- **Desert soil/DG:** Albedo 0.25–0.45 (lighter soils reflect substantially). Lower thermal mass means faster cooling after sunset.
- **Cool roofs (white elastomeric):** Albedo 0.60–0.80. Dramatically reduce roof surface temperatures and building cooling load.

Contribution to UHI Intensity

- Phoenix's UHI intensity is among the strongest in North America, measured at **+7–12°F** for nighttime air temperatures

(urban core vs. rural desert). This is primarily driven by the thermal mass of impervious surfaces slowly releasing stored heat after sunset.
- The massive thermal inertia of asphalt and concrete surfaces delays peak heat release to evening hours, maintaining dangerous temperatures through the night.

Known Cooling Infrastructure

- The City of Phoenix has implemented a **Cool Pavement Pilot Program** using reflective sealcoats on neighborhood streets, which has demonstrated surface temperature reductions of 10–12°F.
- Cool-roof ordinances are encouraged for new commercial buildings. Arizona's energy code promotes reflective roofing materials.
- The city's **Tree and Shade Master Plan** (adopted 2010) targets 25% urban canopy coverage by 2030 (currently estimated at ~10%).
- Shade structures over parking areas and pedestrian corridors have been installed in some public areas (downtown, transit stops).

Sources:

- EPA Urban Heat Island Compendium
- City of Phoenix Cool Pavement Pilot Program reports
- City of Phoenix Tree and Shade Master Plan
- ASU Urban Climate Research Center surface temperature studies
- Oke et al., "Urban Climates" (Cambridge University Press) — surface energy balance reference values

6. Humidity Levels

Typical Relative Humidity Range

- July 15 falls at the onset of the North American Monsoon. Humidity levels in Phoenix can be highly variable during this transition:
  - **Pre-monsoon burst:** Relative humidity at noon can be as low as 5–15% with dew points of 20–40°F.
  - **Active monsoon:** Relative humidity at noon rises to 20–40%, with dew points climbing to 55–65°F, occasionally higher.
- **Estimated for July 15 at noon:** Relative humidity of 15–30%, depending on monsoon moisture status. Typical range for this transitional period.

Dew Point Temperature

- **Typical range:** 40–60°F at this date, reflecting the monsoon moisture influx from the Gulf of California.
- The NWS Phoenix office uses a dew point threshold of 55°F to define the onset of monsoon moisture. Dew points frequently oscillate above and below this threshold in mid-July.
- Even elevated monsoon dew points (55–65°F) remain well below the dangerous heat stress threshold of 75°F, making Phoenix heat primarily a dry heat with lower heat index amplification compared to humid cities.

Humidity & Perceived Temperature

- At the expected actual temperature (~106°F) with a dew point of 50°F (~15% RH), the **Heat Index would be approximately 105–108°F** — close to the actual temperature because low humidity does not amplify perceived heat as significantly.
- However, when monsoon moisture surges push dew points above 60°F at 105°F+ temperatures, the Heat Index can spike to 110–120°F, creating a more dangerous combination than dry heat alone.
- Note: At the anomalous provided temperature of 35°F, humidity is not a heat stress concern, but this temperature is not credible for this location and date.

Seasonal Humidity Patterns

- June is the driest month (average dew point ~20–30°F). The monsoon typically begins in late June to early July, with peak moisture in late July through August.
- The transition from extreme dry heat to humid heat during monsoon onset changes the character of heat stress

significantly — from dehydration/radiation dominance to combined radiation/evaporative stress.

Humidity-Driven Phenomena

- On humid monsoon nights, elevated moisture content traps outgoing longwave radiation more effectively, suppressing nocturnal cooling by 3–5°F compared to dry nights.
- Dew points in Phoenix during monsoon rarely reach the dangerous >75°F threshold, but the combination of 85°F+ nighttime temperatures with elevated moisture creates hazardous conditions for populations without air conditioning.
- Monsoon moisture can paradoxically increase net thermal discomfort despite bringing cloud cover and precipitation, because the reduced evaporative cooling capacity of the body outweighs the benefit of lower peak temperatures.

Sources:

- NOAA/NWS Phoenix dew point and humidity climatology
- NWS Heat Index calculation methodology
- North American Monsoon climatological studies
- Peer-reviewed literature on dry vs. humid heat stress in arid urban environments

7. Solar Radiation Intensity

Expected Global Horizontal Irradiance (GHI)

- Phoenix receives among the highest solar radiation of any U.S. city. In July, clear-sky GHI at solar noon typically reaches **950–1,050 W/m²**.
- Daily total GHI in July averages approximately **7.5–8.5 kWh/m²/day** under clear skies.
- At noon on July 15, under clear or partly cloudy pre-monsoon conditions, estimated GHI is approximately **980–1,020 W/m²**.
- Monsoon cloud buildup (if present) can reduce afternoon GHI by 30–70%, but mornings and early midday are typically clear.

Solar Elevation Angle

- At 33.44°N latitude on July 15 at solar noon (approximately 12:30 PM local standard time / ~1:30 PM MST), the solar elevation angle is approximately **80°** — nearly overhead.
- At this extreme elevation angle, shadows are minimal, vertical surfaces receive reduced direct irradiance, and horizontal surfaces (roads, rooftops, parking lots) receive maximum solar loading.
- The near-vertical sun angle means that shade from buildings and trees only protects areas immediately adjacent to the shade-casting structure, reducing the effectiveness of urban shading compared to lower-latitude seasons.

Peak Sunlight Hours & Thermal Lag

- Sunrise ~5:30 AM, sunset ~7:40 PM — approximately 14 hours of daylight and ~11–12 hours of effective solar heating.
- Solar noon occurs near 12:30 PM MST. However, **peak air temperature typically lags by 3–4 hours**, occurring between 3:00–5:00 PM, as surfaces continue absorbing and re-radiating heat.
- Surface temperatures on asphalt peak between 1:00–3:00 PM, approximately 1–2 hours after solar maximum.

Terrain, Urban Structures & Cloud Cover

- The Phoenix basin is surrounded by low mountain ranges (South Mountain, Camelback, White Tank Mountains) but the urban core sits on flat terrain with no significant topographic shading at midday.
- Low-rise sprawling development (typically 1–2 stories in residential areas) creates minimal canyon shading. Downtown Phoenix has some taller structures (20–40 stories) that create limited street-level shade corridors.
- July cloud cover averages 30–40% (monthly mean), concentrated in afternoon convective development. Morning clear-sky conditions maximize solar loading during the first half of the day.

Differential Surface Absorption

- Under ~1,000 W/m² insolation, dark asphalt absorbs approximately 850–950 W/m² (albedo 0.05–0.15), concrete absorbs 650–800 W/m², light-colored desert soil absorbs 550–750 W/m², and cool/white roofs absorb only 200–400 W/m².
- The differential absorption creates temperature mosaics across the urban landscape, with parking lots and highway corridors acting as concentrated heat sources and irrigated parks serving as thermal relief islands.

Sources:

- NREL National Solar Radiation Database (NSRDB) — Phoenix station data
- NASA Surface Meteorology and Solar Energy (SSE) dataset
- NOAA Solar Calculator for solar elevation angle
- USDA solar radiation modeling for the Sonoran Desert

8. Nighttime Cooling Rates

Diurnal Temperature Range (DTR)

- The typical July DTR in Phoenix is **20–27°F**. With a normal high of ~107°F and low of ~84°F, the range is approximately 23°F.
- This DTR is moderate for a desert climate — natural Sonoran Desert sites outside the urban area experience DTRs of 30–40°F due to rapid radiative cooling of dry, low-thermal-mass surfaces under clear skies.
- The reduced urban DTR compared to the surrounding desert is direct evidence of UHI-suppressed nighttime cooling.

Surface Thermal Mass & Urban Geometry Effects

- Massive heat storage in asphalt, concrete, and masonry surfaces during the day delays and reduces nocturnal longwave radiation loss. The urban fabric acts as a thermal battery, releasing stored heat throughout the night.
- While Phoenix's urban geometry is predominantly low-rise sprawl (low sky view factor obstruction), the sheer extent of impervious surface area more than compensates — the urban footprint is enormous (~500+ square miles of developed land).
- Block walls (concrete masonry units, extremely common in Phoenix residential construction) absorb heat throughout the day and radiate into yards and alleys at night, maintaining elevated near-surface temperatures.

Nighttime Cooling Characterization

- Phoenix exhibits **weak nighttime cooling** in the urban core during July. Overnight temperatures frequently remain above 90°F for extended periods. In 2023, Phoenix recorded 17 consecutive nights that did not drop below 90°F.
- Consecutive hot nights (minimum temperature ≥85°F) create cumulative heat stress because the human body cannot recover adequately during sleep without air conditioning, increasing heat-related mortality risk.
- Cooling rates in the urban core average approximately **2–3°F per hour** in the first few hours after sunset, slowing to less than 1°F/hour after midnight. Rural desert cooling rates can exceed 5°F/hour initially.

Factors Suppressing Nighttime Cooling

- **Thermal mass:** The dominant factor in Phoenix — enormous volume of heat-absorbing impervious surfaces.
- **Anthropogenic heat:** Continuous HVAC exhaust heat (estimated at 20–50 W/m² in dense commercial areas) adds sensible heat to outdoor air throughout the night.
- **Humidity (monsoon):** Elevated water vapor increases atmospheric longwave absorption, reducing net radiative cooling by an estimated 10–20 W/m² on humid nights.
- **Reduced sky view factor:** Even moderate building heights and block walls obstruct the sky view, limiting longwave radiation escape.
- **Calm winds:** Nocturnal wind speeds in the urban core often drop below 5 mph, reducing advective heat removal.

Urban vs. Rural Nighttime Differential

- Nighttime UHI intensity in Phoenix has been measured at **+7–12°F** (urban core vs. rural desert reference), with the strongest differential occurring 3–5 hours after sunset.
- Suburban areas show intermediate values (~+4–7°F above rural), with newer developments sometimes exhibiting stronger UHI due to higher impervious fraction and less mature landscaping.

Sources:

- NOAA/NWS Phoenix diurnal temperature climatology
- ASU Urban Climate Research Center nighttime UHI measurements
- Brazel et al. studies on urban warming and nocturnal temperatures in Phoenix
- Peer-reviewed research on thermal mass effects in arid urban climates

9. Thermal Comfort Index

Applicable Metric Selection

- For Phoenix in July, the **Wet Bulb Globe Temperature (WBGT)** and the **Universal Thermal Climate Index (UTCI)** are the most appropriate metrics.
- **Heat Index** is commonly used by NWS but underestimates the danger of dry heat because it is calibrated for shaded conditions and doesn't fully account for radiant heat load, which is extreme in Phoenix's sun-exposed environments.
- **UTCI** is preferred for a comprehensive assessment because it integrates air temperature, humidity, wind speed, and mean radiant temperature (MRT) — the latter being a critical factor in Phoenix where intense solar radiation and hot surfaces create MRT values of 140–160°F at midday.
- **WBGT** is the most operationally relevant metric for outdoor worker protection, widely used by OSHA and the military, and directly applicable to Phoenix's large outdoor labor force.

Expected Index Values Under Normal July Conditions

- Using expected conditions (air temperature ~107°F, RH ~15–25%, wind speed ~8 mph, full sun exposure):
  - **UTCI: Estimated 110–120°F equivalent** → "Very strong heat stress" to "Extreme heat stress" category
  - **WBGT: Estimated 88–95°F** → Exceeds OSHA recommended work/rest schedules for moderate and heavy work. Black flag condition (military heat stress classification).
  - **Heat Index: ~105–110°F** → NWS "Danger" category; heat cramps and exhaustion likely, heatstroke possible.
- Under the anomalous 35°F reading: UTCI and all thermal comfort indices would indicate **cold stress/slight cold stress**, which is entirely inconsistent with Phoenix in July and strongly suggests data error.

Estimated Comfort Classification (Normal Conditions)

- **"Extreme heat stress"** in direct sun for unacclimatized individuals at midday.
- In shade with wind, conditions moderate to **"strong heat stress"** but remain dangerous for prolonged exposure.
- The extremely high mean radiant temperature (from ground reflection and thermal re-radiation from surfaces) means that shade alone does not eliminate thermal discomfort — radiant heat from surrounding surfaces still contributes significantly.

Heat Stress Risks for Vulnerable Populations

- Maricopa County records the highest number of heat-related deaths in the U.S. — over **645 heat-associated deaths in 2023** alone.
- **Unsheltered/homeless populations:** The most severely affected group. Lack of access to cooling creates life-threatening exposure. Many deaths occur in outdoor settings.
- **Outdoor workers:** Construction, agriculture, landscaping, and delivery workers face extreme occupational heat risk. Arizona lacks mandatory outdoor heat standards (as of early 2024, federal OSHA heat rule was still in development).
- **Elderly and medically vulnerable:** Those with cardiovascular or respiratory conditions face compounded risk. Power outages eliminating AC during heat events are potentially fatal.
- **Children:** Higher surface-area-to-body-mass ratio and closer proximity to superheated ground surfaces increase heat exposure risk.

Local Mitigation Measures

- **Cooling centers:** Maricopa County operates a network of 60+ heat relief stations and cooling centers during the summer season, many at community centers, libraries, and churches.
- **Misting systems:** Deployed at bus stops, outdoor dining areas, commercial walkways, and some public spaces. Common in private commercial venues.

- **Shade structures:** City of Phoenix has invested in shade sails and canopies at transit stops, playgrounds, and pedestrian corridors.
- **Hydration stations:** Free water distribution programs operate during extreme heat warnings.
- **Heat warning system:** NWS Phoenix issues Excessive Heat Warnings (≥113°F expected or 99°F+ overnight lows) which are among the most frequently activated in the nation.

Sources:

- Maricopa County Department of Public Health heat surveillance reports
- OSHA heat illness prevention guidelines
- Jendritzky et al., UTCI methodology documentation
- NWS Heat Index and Excessive Heat Warning criteria
- City of Phoenix Heat Response Plan

10. Seasonal Variability of Urban Heat Islands

UHI Intensity by Season

- **Summer (June–August):** Nighttime UHI intensity is strongest, measured at **+7–12°F** (urban core vs. desert). Daytime UHI is weaker (+2–5°F) because the surrounding desert also heats intensely during the day, reducing the urban-rural contrast.
- **Fall (September–November):** UHI remains significant at **+5–9°F** nighttime as thermal mass retains summer heat and HVAC waste heat remains substantial through October.
- **Winter (December–February):** UHI moderates to **+4–7°F** nighttime. Reduced solar input means less daytime heat storage, but anthropogenic heat (heating systems, traffic) maintains a measurable differential. Winter UHI can actually benefit the city by reducing heating demand.
- **Spring (March–May):** UHI intensifies as solar input increases and days lengthen, reaching **+5–10°F** nighttime by May. The onset of intensive HVAC cooling adds anthropogenic heat.

Driving Factors

- **Foliage cycles:** Limited deciduous vegetation in Phoenix (most landscaping is evergreen desert species or palms). Mesquite and palo verde trees leaf out in spring, providing some seasonal canopy variation, but the effect on UHI is modest compared to temperate cities.
- **Monsoon effects (July–September):** Increased cloud cover and moisture during monsoon can slightly reduce daytime UHI by limiting solar loading, but increased humidity can enhance nighttime UHI by trapping longwave radiation. Net effect is complex and event-dependent.
- **No snow/ice influence:** Phoenix essentially never receives snow. There is no seasonal albedo change from snow cover, unlike northern cities where UHI is maximized in winter (dark surfaces vs. snow-covered rural areas).
- **HVAC load:** Summer cooling demand (residential AC runs nearly continuously from June–September) generates massive anthropogenic waste heat. Estimated at 30–60 W/m² in commercial districts. This is a dominant summertime UHI driver unique in magnitude among U.S. cities.
- **Irrigation seasonality:** Landscape irrigation peaks in summer, providing localized evaporative cooling in irrigated areas. Reduced irrigation in winter slightly changes the moisture landscape.

Urban-Rural Temperature Differentials Across Seasons

- The rural reference (open Sonoran Desert) experiences much larger seasonal and diurnal temperature swings than the urban core:
  - Summer: Rural highs ~105–110°F, lows ~72–78°F (DTR ~30–35°F). Urban lows ~84–90°F (DTR ~20–23°F).
  - Winter: Rural lows can drop to ~30–35°F on clear nights while urban core stays ~40–45°F.
- The moderating (buffering) effect of urban thermal mass is evident year-round but most consequential for health in summer when elevated nighttime temperatures prevent physiological recovery.

Seasonal Adaptation Measures

- **Summer-specific:** Cooling centers activated May–September. Heat emergency declaration protocols. Public awareness campaigns. Work schedule modifications (early start times for outdoor construction, typically 5:00 AM).
- **Year-round infrastructure:** Cool pavement programs (ongoing expansion), tree planting campaigns, building energy

code updates promoting cool roofs and insulation.

- **Planned:** The City of Phoenix appointed the nation's first publicly funded municipal **Chief Heat Officer** in 2021, coordinating comprehensive heat mitigation strategy. The HeatReady initiative focuses on long-term resilience investments including expanded green infrastructure and shade equity mapping.
- Maricopa County's climate action plan includes heat-specific zoning recommendations (minimum shade requirements for parking lots and pedestrian areas in new developments).

Season of Greatest Compound Heat Risk

- **Late June through mid-July** poses the greatest compound heat risk, specifically the pre-monsoon to early monsoon transition period (which includes the date in question, July 15).
- This period combines: (1) maximum solar elevation and day length, (2) peak temperatures frequently exceeding 115°F, (3) variable humidity (dry conditions exacerbate dehydration; sudden monsoon moisture spikes amplify heat index), (4) maximum UHI intensity from cumulative seasonal heat storage, (5) peak ozone and air quality degradation, and (6) monsoon dust storms (haboobs) that can rapidly degrade air quality.
- The compounding of extreme heat with air quality events, potential power grid strain (peak demand periods), and episodic humidity surges makes this the most dangerous period for heat-related morbidity and mortality in the Phoenix metropolitan area.

Sources:

- ASU Urban Climate Research Center seasonal UHI studies
- Brazel et al., seasonal and diurnal UHI dynamics in Phoenix
- City of Phoenix Office of Heat Response and Mitigation
- Maricopa County Department of Public Health heat mortality surveillance data
- EPA Heat Island Effect documentation and reduction strategies
- NWS Phoenix climatological summary and monsoon guidance

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